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# DESIGN AND ANALYSIS OF ALGORITHM

## Finding time complexity

Question 1:

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void function (int n)
{
   int i= 1;
   int s =1;
   while(s <= n)
   {
       i++;
       s += i;
    }
}
```

**Note:** No need of counter increment for declarations and scanf() and count variable printf() statements.

**Input:**
A positive Integer n

**Output:**
Print the value of the counter variable

**For example:**

| Input | Result |
|---|---|
| 9 | 12 |

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Answer:

```
#include <stdio.h>

int count = 0;

void function(int n){
    int i = 1;
    count++;
    int s = 1;
    count++;
    while(s <= n){
        count++;
        i++;
        count++;
        s+=i;
        count++;
    }
    count++;
}

int main(){
    int n;
    scanf("%d", &n);
    function(n);
    printf("%d", count);
    return 0;
```

```
}
```

Question2:

Convert the following algorithm into a program and find its time complexity using the counter method.

```
void func(int n)
{
   if(n==1)
   {
     printf("*");
   }
   else
   {
    for(int i=1; i<=n; i++)
    {
      for(int j=1; j<=n; j++)
      {
         printf("*");
         printf("*");
         break;
      }
    }
  }
}
```

**Note:** No need of counter increment for declarations and scanf() and count variable printf() statements.

**Input:**

A positive Integer n

**Output:**

Print the value of the counter variable

**Answer:**

```
#include <stdio.h>


int count = 0;


void func(int n){
   if(n==1){
      count++;
      //printf("*");
      count++;
   }
   else{
      count++;
      for(int i = 1; i <= n; i++){
         count++;
         for(int j = 1; j <= n; j++){
            count++;
            //printf("*");
            count++;
            //printf("*");
            count++;
            break;
         }
```

```
            count++;
        }
        count++;
    }
}


int main(){
    int n;
    scanf("%d", &n);
    func(n);
    printf("%d", count);
    return 0;
}
```

Question 3:

Convert the following algorithm into a program and find its time complexity using counter method.

```
 Factor(num) {
 {
    for (i = 1; i <= num;++i)
    {
     if (num % i== 0)
        {
          printf("%d ", i);
        }
     }
  }
```

**Note:** No need of counter increment for declarations and scanf() and counter variable printf() statement.

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**Input:**

A positive Integer n

**Output:**

Print the value of the counter variable

**Answer:**

```
#include <stdio.h>


int count = 0;


void Factor(int num){
    for(int i = 1; i <= num; i++){
        count++;
        if(num%i==0){
            //printf("%d", i);
            count++;
        }
        count++;
    }
    count++;
}


int main(){
    int n;
    scanf("%d", &n);
    Factor(n);
    printf("%d", count);
    return 0;
```

```
}
```

Convert the following algorithm into a program and find its time complexity using counter method.

```
void function(int n)
{
   int c= 0;
   for(int i=n/2; i<n; i++)
      for(int j=1; j<n; j = 2 * j)
         for(int k=1; k<n; k = k * 2)
            c++;
}
```

**Note:** No need of counter increment for declarations and scanf() and count variable printf() statements.

**Input:**
A positive Integer n
**Output:**
Print the value of the counter variable

Answer:

```
#include <stdio.h>

int count = 0;

void function(int n){
   int c = 0;
   count++;
   for(int i = n/2; i < n; i++){
      count++;
```

```
        for(int j = 1; j < n; j = 2 * j){
            count++;
            for(int k = 1; k < n; k = k * 2){
                count++;
                c++;
                count++;
            }
            count++;
        }
        count++;
    }
    count++;
}

int main(){
    int n;
    scanf("%d", &n);
    function(n);
    printf("%d", count);
    return 0;
}
```

Question 5:

Convert the following algorithm into a program and find its time complexity using counter method.

```
void reverse(int n)
{
  int rev = 0, remainder;
  while (n != 0)
```

```
   {
      remainder = n % 10;
      rev = rev * 10 + remainder;
      n/= 10;

   }
print(rev);
}
```

**Note:** No need of counter increment for declarations and scanf() and count variable printf() statements.

**Input:**
A positive Integer n
**Output:**
Print the value of the counter variable

**Answer:**

```
#include <stdio.h>


int count = 0;


void reverse(int n){
   int rev = 0, remainder;
   count++;
   while(n != 0){
      count++;
      remainder = n%10;
      count++;
      rev = rev * 10 + remainder;
      count++;
      n /= 10;
```

```
        count++;
    }
    count++;
    //print(rev);
    count++;
}

int main(){
    int n;
    scanf("%d", &n);
    reverse(n);
    printf("%d", count);
    return 0;
}
```